

## DSA8045 – Applied Analytics Assignment 1 datasets

### Group 1 – Movies

#### Dataset Overview

This dataset contains information about movies, including ratings, critics' reviews, release information and award nominations.

Variables to use:

- **title** – Title of the movie.
- **title\_type** – Type of movie.
- **genre** – Primary genre of the movie.
- **runtime** – Length of the movie in minutes.
- **mpaa\_rating** – MPAA age classification rating.
- **thtr\_rel\_year** – Year the movie was released in cinemas.
- **thtr\_rel\_month** – Month the movie was released in cinemas.
- **thtr\_rel\_day** – Day of the month the movie was released in cinemas.
- **imdb\_rating** – Average IMDb rating awarded by users.
- **imdb\_num\_votes** – Number of IMDb users who voted for the movie.
- **critics\_rating** – Categorical Rotten Tomatoes critics rating (e.g., Fresh or Rotten).
- **critics\_score** – Rotten Tomatoes critics score (0–100).
- **best\_pic\_nom** – Indicates whether the movie was nominated for an Academy Award for Best Picture.
- **best\_pic\_win** – Indicates whether the movie won the Academy Award for Best Picture.
- **best\_actor\_win** – Indicates whether one of the lead actors has won an Academy Award.
- **best\_actress\_win** – Indicates whether one of the lead actresses has won an Academy Award.
- **best\_dir\_win** – Indicates whether the director has won an Academy Award.
- **top200\_box** – Indicates whether the movie appeared in the Top 200 Box Office list.
- **director** – Name of the movie director.

## Group 2 – AirBnB Listings

### Dataset Overview

This dataset contains information about Airbnb properties including location, pricing, room type, availability and customer reviews.

Variables to use:

- **id** – Unique identifier for the Airbnb listing.
- **name** – Name/title of the Airbnb listing.
- **host\_id** – Unique identifier for the host.
- **host\_name** – Name of the host.
- **neighbourhood** – Specific neighbourhood where the property is located.
- **latitude** – Latitude coordinate of the property.
- **longitude** – Longitude coordinate of the property.
- **room\_type** – Type of accommodation offered.
- **price** – Price per night.
- **minimum\_nights** – Minimum number of nights required for booking.
- **number\_of\_reviews** – Total number of reviews received.
- **last\_review** – Date of the most recent review.
- **reviews\_per\_month** – Average number of reviews received each month.
- **calculated\_host\_listings\_count** – Number of Airbnb listings managed by the host.
- **availability\_365** – Number of days the property is available during the year.
- **number\_of\_reviews\_ltm** – Number of reviews received during the last 12 months.

## Group 3 – Spotify Music

### Dataset Overview

This dataset contains information about songs and their audio characteristics obtained from Spotify. It includes measures of popularity, musical features, artist information and genre classifications.

Variables to use:

- **ID** – Unique identifier for the track.
- **artists** – Name of the artist or artists who performed the track.
- **album\_name** – Name of the album on which the track appears.
- **track\_name** – Name of the song.
- **popularity** – Spotify popularity score ranging from 0 to 100, where higher values indicate more popular tracks.
- **duration\_ms** – Length of the track in milliseconds.
- **explicit** – Indicates whether the song contains explicit content.
- **danceability** – Measure of how suitable a track is for dancing (0–1). Higher values indicate a stronger dance rhythm.
- **energy** – Measure of intensity and activity (0–1). High-energy tracks are typically fast, loud and energetic.
- **key** – Estimated musical key of the track (represented numerically).
- **loudness** – Average loudness of the track measured in decibels.
- **mode** – Musical modality of the track. 1 = Major key and 0 = Minor key.
- **speechiness** – Measure of the presence of spoken words in a track (0–1).
- **acousticness** – Confidence measure of whether the track is acoustic (0–1).
- **instrumentalness** – Measure of the likelihood that the track contains no vocals (0–1).
- **liveness** – Measure of the likelihood that the track was recorded live (0–1).
- **valence** – Measure of musical positivity or happiness (0–1). Higher values indicate more positive or cheerful songs.
- **tempo** – Estimated tempo of the track measured in beats per minute.
- **time\_signature** – Estimated number of beats per musical bar.
- **track\_genre** – Genre classification assigned to the track.

## Group 4 – Store Sales

### Dataset Overview

This dataset contains sales transactions from a retail store. Each row represents an individual customer order and includes information about products, customers, locations, shipping methods, sales revenue, discounts and profit.

Variables to use:

- **Order.ID** – Unique identifier for each customer order.
- **Order.Date** – Date the order was placed.
- **Ship.Date** – Date the order was shipped.
- **Ship.Mode** – Shipping method used to deliver the order.
- **Customer.ID** – Unique identifier for the customer.
- **Customer.Name** – Name of the customer.
- **Segment** – Customer segment (Consumer, Corporate, Home Office).
- **Country** – Country where the customer is located.
- **City** – City where the customer is located.
- **State** – State or region where the customer is located.
- **Postal.Code** – Postal or ZIP code associated with the customer.
- **Region** – Sales region (e.g., East, West, Central, South).
- **Product.ID** – Unique identifier for the product.
- **Category** – Main product category.
- **Sub.Category** – More detailed product classification within a category.
- **Product.Name** – Name of the product.
- **Sales** – Revenue generated from the sale in US dollars.
- **Quantity** – Number of units sold.
- **Discount** – Discount applied to the sale.
- **Profit** – Profit generated from the sale in US dollars.
- **Profit.Margin** – Profit margin expressed as a proportion or percentage of sales.

## Group 5 – Coffee Quality

### Dataset Overview

This dataset contains information on coffee samples evaluated by professional coffee tasters. Each observation represents a coffee sample and includes details about its origin, production methods, physical characteristics and quality scores.

Variables to use:

- **Species** – Species of coffee bean being evaluated.
- **Country.of.Origin** – Country where the coffee was produced.
- **Region** – Region within the country where the coffee was grown.
- **Producer** – Producer, farm or organisation responsible for the coffee.
- **Number.of.Bags** – Number of bags included in the coffee lot.
- **In.Country.Partner** – Organisation responsible for grading, certifying or evaluating the coffee within the producing country.
- **Harvest.Year** – Year in which the coffee was harvested.
- **Grading.Date** – Date on which the coffee was evaluated and graded.
- **Processing.Method** – Method used to process the coffee beans after harvesting.
- **Aroma** – Score awarded for the coffee's aroma.
- **Flavor** – Score awarded for the flavour quality of the coffee.
- **Aftertaste** – Score awarded for the quality and persistence of flavour after consumption.
- **Acidity** – Score awarded for the coffee's acidity and brightness.
- **Body** – Score awarded for the texture and mouthfeel of the coffee.
- **Balance** – Score awarded for how well the coffee's flavour characteristics work together.
- **Uniformity** – Score awarded for consistency across multiple cups of the same coffee.
- **Clean.Cup** – Score awarded for the absence of defects or undesirable flavours.
- **Sweetness** – Score awarded for the perceived sweetness of the coffee.
- **Cupper.Points** – Additional points awarded by professional coffee tasters.
- **Total.Cup.Points** – Overall quality score assigned to the coffee sample.
- **Moisture** – Moisture content of the coffee beans.
- **altitude\_low\_meters** – Lowest altitude at which the coffee was grown (metres above sea level).
- **altitude\_high\_meters** – Highest altitude at which the coffee was grown (metres above sea level).
- **altitude\_mean\_meters** – Average altitude at which the coffee was grown (metres above sea level).